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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 09

B.Tech. (CE) (Sem.–6)

BRIDGE ENGINEERING

Subject Code : PECE-603-F-18

M.Code : 79405

Date of Examination: 19-12-2022

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

SECTION-A

1. Write briefly :

a) Mention various site selection factors for concrete bridge.

b) What do you mean by optimum span of bridge?

c) What are the advantages of elastomeric bearing over other types of bearings?

d) Write down the function of tower for a cabal stayed bridge.

e) Draw typical cross-section shapes of piers for river and waterways crossing.

f) What do you mean by cast-in-situ construction?

g) What is a deep foundation?

h) How is contact pressure distributed for a ridge footing?

i) What is an arch-bridge?

j) When is expansion joints provided in bridge?

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SECTION-B

2. What is the procedure that may be used for selection of the bearing?

3. Discuss various types of abutments with their uses.

4. Discuss in detail design procedure for a plate girder bridge.

5. What are the advantages and disadvantages of cantilever bridge.

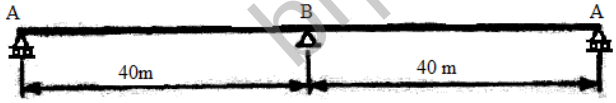
6. Elaborate geometric and hydraulic design consideration for designing a bridge.

SECTION-C

7. What are the various special requirements for prestressed concrete bridges? Elaborate

8. What are various forces acting on a well foundation? Discuss in detail

9. Design an elastomeric bearing with the help of given data
   L = expandable span length = 40 m
   R_{DL} = DL reaction/ girder = 690 kN
   R_{LL} = II reaction (without impact)/girder = 220 kN
   $\square_S$ = bearing design rotation at service limit state = 0.0025 rad
   ΔT maximum temperature change = 21-degree celcius
   Δ_{PT} = girder shortening dure to concrete shrinkage = 2 mm
   G = shear modulus of elastomer = 0.9 to 1.38 MPa
   y = load factor for uniform temperature = 1.2
   ΔF_{TH} = constant amplitude fatigue threshold for Category A = 165 MPa
   Using 60 durometer reinforced bearing:
   Fy = yield strength of steel reinforcement = 350 MPa
   Use slidifig bearing,
   (Assume data, if required)



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